

Exact Selector-Threshold Matching and Nonlinear Momentum-Lattice Cascade

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral; see RECOVERY_PROVENANCE.md.

45. Factorised three-loop matching

Because the first mixed graph is two-particle reducible, its zero-momentum coefficient factorises into a one-loop selector kernel and a mass derivative of the known two-loop fermion-fermion-scalar effective potential [Mar-tin2001]. In the scalar proxy,

$$\mathcal{J}_3(\bar{\mu}) = 2m_R^2 K_{RRH}(m_R^2, m_h^2) D_{FFS}(M^2, m_h^2; \bar{\mu}).$$

At $\bar{\mu} = M$,

$$\mathcal{J}_3(M) = 6.57973508149 \simeq \frac{2\pi^2}{3}.$$

The first proxy coefficient gave

$$|\Delta V_{Qa}^{(3)}| = 2.1723 \times 10^3 \text{ GeV}^4,$$

or only 1.546×10^{-12} of the intended thermal focusing potential.

The fixed-order hard function had a huge apparent scale excursion. This was explicitly identified as missing RG completion, not a physical uncertainty.

46. Spurion-graded operator mixing

A correction to the earlier claim was necessary. Exact Z_6 does not make the complete invariant transient basis block diagonal, because combinations $e^{ipx}\mathcal{X}_{-p}$ are individually invariant. The correct selection rule is

$$\gamma_{(A,p)(B,q)} = \sum_{r,s \geq 0} \delta_{p-q,r-s}^{(6)} \epsilon^{r+s} \Gamma_{AB}^{(r,s)}.$$

Changing harmonic charge requires explicit shift-breaking spurions. The powers of ϵ are fixed, although the finite tensors depend on the messenger completion.

47. Nonlinear momentum-lattice cascade and corrected branching

The repaired sequence is

$$\phi \rightarrow N_0 \bar{N}_0 \rightarrow R_0 \nu_0 \rightarrow H_0, H_5 \rightarrow D_k, q_k, g_k.$$

A radial momentum-lattice calculation included expansion, backreaction, two-body decays, Pauli blocking, and an energy-conserving plasma closure. It exposed a genuine error in the v0.9 branching argument.

The massless ν_0 daughter deposits visible-sector energy earlier than the reheaton products and experiences a different redshift history. Therefore the attractive exact choice

$$\tan \theta = \frac{1}{16}$$

does not yield the desired final temperature ratio. The simulation gave

$$\mathcal{R}_\nu = \frac{E_\nu}{E_R^{(1)}} = 0.35551328.$$

Imposing $E_5/E_0 = 1/256$ requires

$$B_5^* = \frac{1 + \mathcal{R}_\nu}{257} = 0.00527437074,$$

The v1.2 note reported

$$\tan \theta = 0.0728196.$$

Direct recomputation from the displayed value of B_5^* gives

$$\tan \theta = \sqrt{\frac{B_5^*}{1 - B_5^*}} = 0.07281715.$$

The difference is a small historical rounding inconsistency and is recorded rather than silently erased. The resulting v1.2 calculation obtained

$$\frac{T_5}{T_0} = 0.250000001.$$

After restoring the complete Higgs-doublet and relativistic- D contribution to g_* in v1.3, the superseding branch was

$$B_5 = 0.00529888708, \quad \tan \theta = 0.0729871.$$

This correction is retained as a warning against replacing a resolved cascade with an aesthetically neat branching identity.